

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	<i>Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2</i>		
Student Name:	TEJASWINI . M	Reg. No.:	192411079
Section / Batch:	CSE - 2024 BATCH	Date:	15/07/26

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
GRAND TOTAL:			100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

- Draw a snowflake schema diagram for the data warehouse.
- Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?
- If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.

(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.

(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:

Product: Category, type, and batch.

Factory: Location, department, and line number.

Time: Year, quarter, month, and week.

Supplier: Region, rating, and contract type.

- Measures:

Defect rate.

Production units.

Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:

Product: Category, subcategory, and SKU.

Store: Region, city, and store ID.

Time: Year, quarter, month, and day.

Customer: Membership tier, age group, and gender.

- Measures:

Inventory turnover.

Total sales revenue.

Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

Code of Conduct

I Tejaswini M (192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M. Tejaswini

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks)							
Overall Total (100)							

Faculty In-charge

Course Coordinator

Head of Department

Signature: _____

Signature: _____

Signature: _____

Date: _____

Date: _____

Date: _____

CSAIGIO - DATA WAREHOUSING AND DATA MINING

COI - ASSESSMENT TOOL - I
SHORT ANSWERS TEST

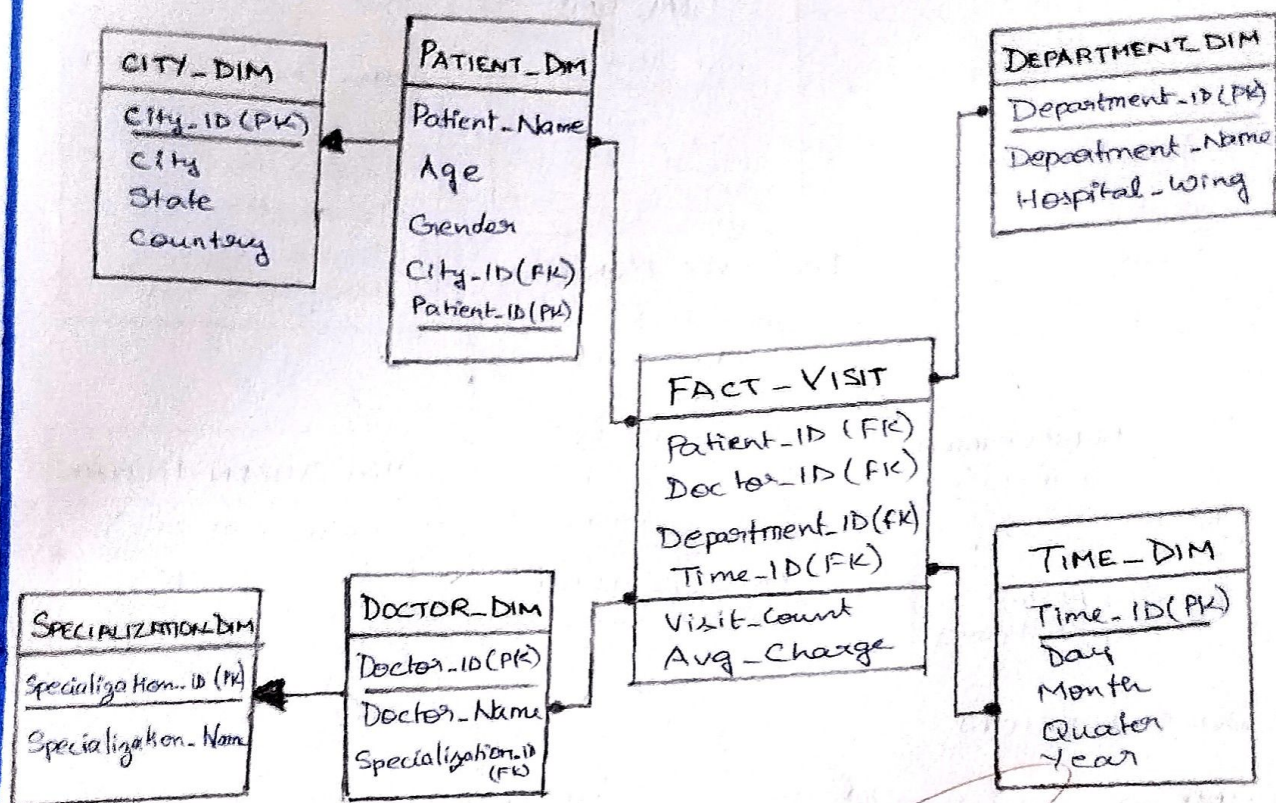
TEJASWINI. M

192411079

BE CSE

1) CITY HOSPITAL DATA WAREHOUSE

a) SNOWFLAKE SCHEMA DIAGRAM



b) OLAP OPERATIONS

To find average consultation fee department-wise for each year:

- ROLL-UP PATIENT $\rightarrow$ ALL
- ROLL-UP DOCTOR $\rightarrow$ ALL
- ROLL-UP TIME $\rightarrow$ YEAR
- Keep Department unchanged

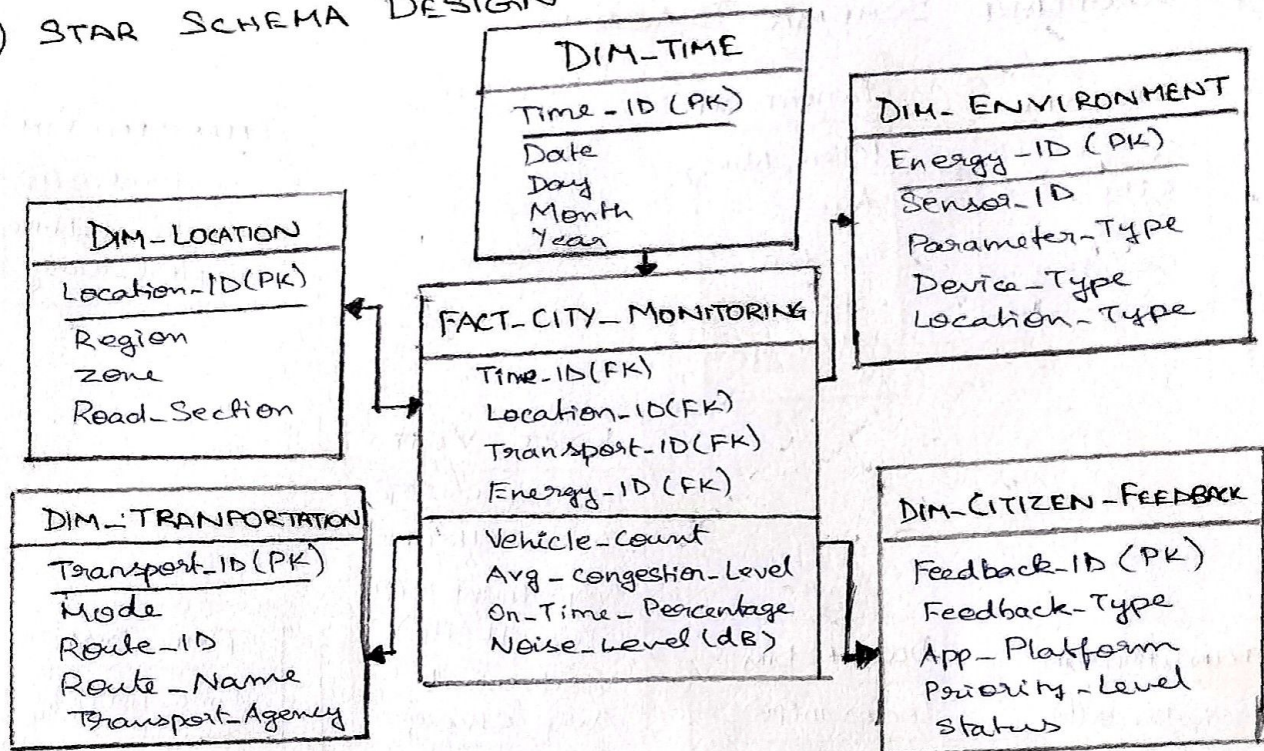
RESULT: Average consultation fee for each department in every year.

c) TOTAL CUBOIDS

$$\begin{aligned} \text{Formula} &= L_1 \times L_2 \times L_3 \times L_4 \\ &= 4 \times 4 \times 4 \times 4 \\ &= 256 \text{ cuboids} \end{aligned}$$

2) SMART CITY PROJECT

a) STAR SCHEMA DESIGN

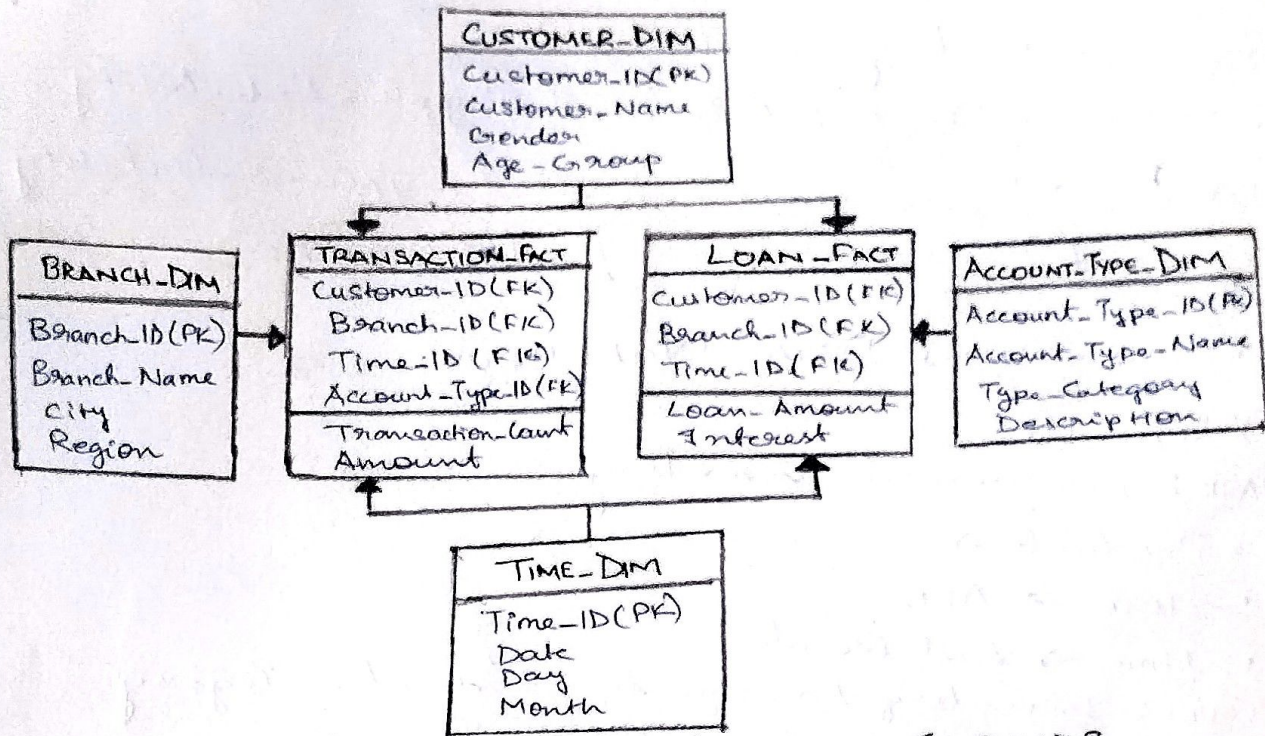


b) JUSTIFICATION

- **TIME** → Analyzes daily, monthly, and yearly trends
- **LOCATION** → Compares different city areas
- **TRAFFIC** → Monitors vehicle movement and congestion
- **ENVIRONMENT** → Tracks air quality, noise and energy usage
- **TRANSPORTATION** → Measures public transport efficiency
- **CITIZEN FEEDBACK** → Evaluates public satisfaction and service quality

2) BANKING DATA WAREHOUSE

a) GALAXY SCHEMA



b) OLAP Operations

To obtain branch-wise average transaction

- Roll-Up Time → Year
- Keep Branch dimension
- Aggregate :
Avg (Transaction Amt)
Avg (Loan Amt)

c) TOTAL CUBOIDS

Formula = Levels⁴

$$= 4 \times 4 \times 4 \times 4$$

$$= 256$$

256 cuboids (including base and apex cuboids)

4) MANUFACTURING COMPANY DATA WAREHOUSE

a) ROLL-UP

Product → Category

Factory → ALL

Time → Current Year

Defect rate by product category

b) DRILL-DOWN

Select Factory

Drill down → Department

Product → Type

Defect rate by department and product type

c) SLICE

Supplier Rating = 4+

RESULT:

Production units for suppliers rated 4 or above

d) DICE

- Select Product Categories
- Select Factory Locations
- Measures: Defect Rate and Supplier Reliability

RESULT: Compare defect rate and supplier reliability for selected categories and factories.

5) RETAIL CHAIN DATA WAREHOUSE

a) ROLL-UP

TASK: Summarize inventory turnover

- Product → Category
- Store → ALL
- Time → Last Quarter

RESULT: Inventory turnover by product category

b) DRILL-DOWN

TASK: Analyze electronics sales by subcategory & SKU

- Product → ELECTRONICS → SUBCATEGORY → SKU -
- Store → SPECIFIC REGION

RESULT: Sales data by subcategory and SKU

c) SLICE

TASK: Isolate sales revenue for male customers with Bronze membership

- Customer → Male
- Membership Type → Bronze

RESULT: Sales revenue for male Bronze customers

d) DICE

TASK: Analyze sales revenue and units sold for selected cities, product categories and time period

- Select Cities
- Select Product Categories

MEASURES: Sales Revenue, Units Sold.